HIP-3/HIP-4 Symbiosis: How can HIP-4 enhance HIP-3

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Abstract

HIP-3 enables permissionless perpetual futures on Hyperliquid with bounded mark prices and asset-specific discovery bounds. HIP-4 introduces fully collateralised binary outcome contracts with one-block settlement. This paper studies their interaction within a unified margin account through four experiments combining documented protocol mechanics [DOC], empirical calibration from 449 observed weekends [EMP], and Monte Carlo simulation [SIM].

The core argument is that HIP-3's bounded mark-price mechanism creates two risk channels (gap risk at weekend reopen, funding variance during the internal pricing session) that HIP-3 alone cannot hedge, and that HIP-4 provides the missing instruments from within the same margin account. Four findings support this. The relayer clamp creates an exploitable settlement window that HIP-4's block-finality eliminates [DOC]. Portfolio margin recognition yields 7–14% capital efficiency gains in cross-margin markets [SIM]. Binary put hedges eliminate bad debt at the Level 0 bound and reduce it 50–99% beyond [SIM]. Funding-binary fair values range from 1.6% to 16.6% across regime assumptions, providing a pricing framework for an instrument that does not yet have liquid markets [SIM]. A counterparty analysis and governance proposal follow.

Simulation results are sensitive to model assumptions; quantitative magnitudes are indicative. Protocol parameters from trade.xyz and Hyperliquid documentation as of April 2026. Weekend gap distributions from Yahoo Finance, 2018–2025.

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1 Introduction

Hyperliquid's HIP-3 allows permissionless deployment of perpetual futures by any builder staking 500,000 HYPE. Trade.xyz uses this to operate equity, commodity, and index perps that trade continuously through weekends. HIP-4, on testnet since February 2026, adds fully collateralised binary outcome contracts with block-finality settlement.

Consider a trader holding a $100k WTIOIL perpetual at 10× leverage through a weekend on trade.xyz. HIP-3's bounded mark-price mechanism exposes this position to two risks with no on-chain hedge:

1. **Gap risk.** Discovery Bounds v2 allows the mark price to ratchet up to ±15.76% from the Friday close through sequential re-anchoring. If the Monday open exceeds the liquidation threshold (−10% at 10×), the position is insolvent.

2. **Funding variance.** The internal EWMA oracle and 0.5×-scaled funding formula generate continuous payments for 48 hours. A sustained crowd imbalance can accumulate material funding costs even if the Monday open gap is small.

HIP-4 addresses both. A *price binary* struck at the liquidation threshold pays out if the gap exceeds it, covering the perp loss (Experiment 3). A *funding binary* above a cumulative-funding threshold pays out if weekend funding exceeds a target, capping the trader's cost (Experiment 4). Both instruments settle within the same margin account, enabling portfolio margin recognition that frees collateral (Experiment 2). Capital efficiency can be partially approximated off-chain at additional counterparty cost and latency; gap hedging and funding-variance hedging depend on shared on-chain state and atomic execution.

1.1 Evidence Classification

Results are tagged by evidence type. [DOC]: derived mechanically from documented protocol parameters. [EMP]: calibrated from observable data (Yahoo Finance gaps); carries sampling uncertainty and proxy risk. [SIM]: produced by simulation with assumed parameters; sensitive to model specification.

2 Parameters and Assumptions

All parameters are from trade.xyz[1] and Hyperliquid[2] documentation unless stated otherwise.

[1] https://docs.trade.xyz, accessed March 2026.
[2] https://hyperliquid.gitbook.io/hyperliquid-docs, accessed March 2026.

2.1 Notation

Symbol	Definition
$S(t)$, $F(t)$	Internal oracle price, perpetual mark price
$\pi(t)$	Outcome contract price (implied probability) $\in [0, 1]$
K, T	Strike threshold, expiry of outcome contract
N_p, N_o	Perp and outcome notional
x_p, $C_p = N_p/x_p$	Leverage, perpetual collateral
$\mathcal{F}$	Cumulative weekend funding (% of notional)

2.2 Oracle and Funding

When external markets are closed, the XYZ oracle advances the internal price via an exponentially weighted moving average of the impact price difference (IPD):

$$S_t = \beta_t S_{t^-} + (1 - \beta_t)(S_{t^-} + \mathrm{IPD}_t), \quad \beta_t = e^{-\Delta t^*/\tau}, \quad \Delta t^* = \min(\Delta t,\, c\tau), \tag{1}$$

with time constant $\tau = 1\,\mathrm{h}$ and cap $c = 0.1$, so each update moves the oracle by at most $1 - e^{-0.1} \approx 9.5\%$ of the IPD. The XYZ funding rate applies a $0.5\times$ multiplier to the standard Hyperliquid formula:

$$F_{\mathrm{XYZ}}(t) = 0.5\big[P(t) + \mathrm{clamp}(r - P(t),\, -0.05\%,\, +0.05\%)\big], \tag{2}$$

where $P(t) = \mathrm{IPD}(t)/S(t)$ is the premium index (sampled every 5 s, hourly averaged), $r = 0.000625\%/\mathrm{h}$ is the scaled interest rate, and the cap is $\pm 4\%/\mathrm{h}$. Funding is paid on the oracle price, not the mark price.

2.3 Discovery Bounds v2

Asset	Bound	Max resets
SP500	±2%	1
WTIOIL	±5%	2
GOLD, SILVER	±4%	1–2
Most equities	±10%	0

At 90% of the bound distance, the reference price re-anchors and a fresh range opens. WTIOIL's maximum discoverable range is $1.05^3 - 1 \approx 15.76\%$.

2.4 Empirical Gap Distribution

To calibrate the gap scenarios in Experiment 3, we fit weekend gap return distributions to Yahoo Finance 2018–2025 data (449 weekends per asset). Each asset's distribution is split at threshold $= b/2$ into a normal component and a jump component [EMP]:

Asset	n	σ_{norm}	p_{jump}	95% CI	$P(g < -b)$	w_{crisis}
SPY (b=±2%)	454	0.43%	14.1%	[11.0,17.4]%	1.76%	2.9%
CL=F (b=±5%)	449	0.59%	4.0%	[2.4,5.8]%	0.22%	0.9%
TSLA (b=±10%)	454	1.88%	6.2%	[4.0,8.6]%	0.88%	1.3%

Threshold $= b/2$. These are Yahoo Finance proxies; XYZ on-chain distributions may differ due to the bounded mark-price mechanism.

3 Experiment 1: Settlement Arbitrage Window

Before analysing composability, we establish why HIP-4's settlement design is a prerequisite. The HIP-3 relayer clamps mark price updates at ±50 bps per tick (≈3 s intervals). A binary contract deployed as a HIP-3 perpetual must therefore migrate from $\pi(t^-)$ to $\{0, 1\}$ through sequential 50 bps ticks. Ticks to settle: $n = \lceil (1 - \pi(t^-))/0.005 \rceil$. Calendar window: $T = 3n$ seconds. Extractable arbitrage:

$$V_{\text{arb}} = \sum_{k=0}^{n-1} [1 - (\pi(t^-) + 0.005k)] \times 0.005. \tag{3}$$

At $\pi(t^-) = 0.5$: $T = 300$ s, $V_{\text{arb}} \approx 12.6\%$ [DOC]. Binary settlement is incompatible with the relayer-clamp oracle; HIP-4's block-finality resolution (≈0.4 s) eliminates it. If HIP-4 requires an additional oracle round (≈3 s), the advantage narrows to 100× but does not disappear. *Evidence:* mechanical [DOC].

4 Experiment 2: Capital Efficiency Under Portfolio Margin

Scope: Cross-margin HIP-3 markets. Most XYZ markets enforce `strictIsolated` margin; the seven MAG7 equity perps (GOOGL, AMZN, AAPL, META, MSFT, NVDA, TSLA) support cross-margin since February 2026. This experiment applies directly to MAG7 markets and prospectively to others pending margin-mode extension (see Governance Proposal).

A long perpetual and a long binary put on the same underlying have opposite delta exposure: a falling price hurts the perp and increases the put's value. A margin engine that recognises this offset can release the collateral locked as duplicate downside protection.

$N = 2{,}000$ paths ($\mu = 0$, $\sigma = 2\%$/h, 24 h), long perp ($N_p = \$100$k, $x_p = 10$) hedged with a binary put at $\pi_0 \in \{5\%, \ldots, 35\%\}$, $K = 90\%$, fixed hedge notional $N_o = C_p$. The margin engine evaluates the put's expected coverage at each hourly price along the path:

$$\text{coverage}_t = \min\big(1,\ \pi_{\text{put}}(S_t) \cdot (1 - \pi_0)\big), \tag{4}$$

where $\pi_{\text{put}}(S_t)$ is the binary put's mark-to-market price at the current oracle price. The margin offset Ω is the time-averaged coverage times C_p, and capital efficiency is $\eta = \Omega/(C_p + \pi_0 N_o) \times 100\%$.

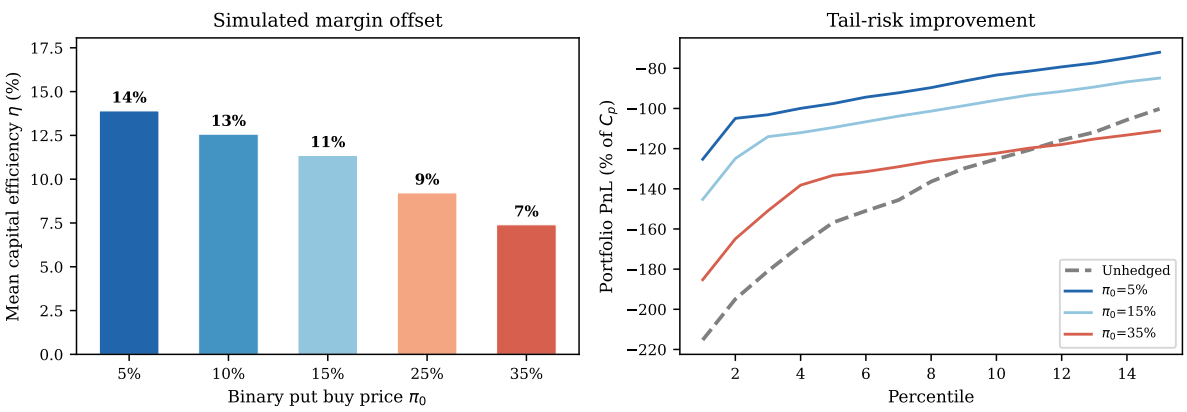

Figure 1: Capital efficiency [SIM]. *Left:* Mean η by buy price: 7% ($\pi_0 = 35\%$) to 14% ($\pi_0 = 5\%$). *Right:* Tail-risk improvement vs. unhedged (dashed). $N = 2{,}000$ paths.

Evidence: simulated [SIM]; contingent on undocumented binary delta recognition and cross-margin mode.

5 Experiment 3: Weekend Gap Hedging

Simulated 48-hour weekends for three assets under Discovery Bounds v2 gap scenarios:

Asset	Bound	Resets	Scenario A	B	C
SP500	±2%	1	−2%	−4.04%	−5.5%
WTIOIL	±5%	2	−5%	−10.25%	−15.76%
TSLA	±10%	0	−10%	−15%	−20%

Binary put hedges at $\pi_0 \in \{5\%, 10\%, 20\%\}$ deployed at the liquidation threshold.

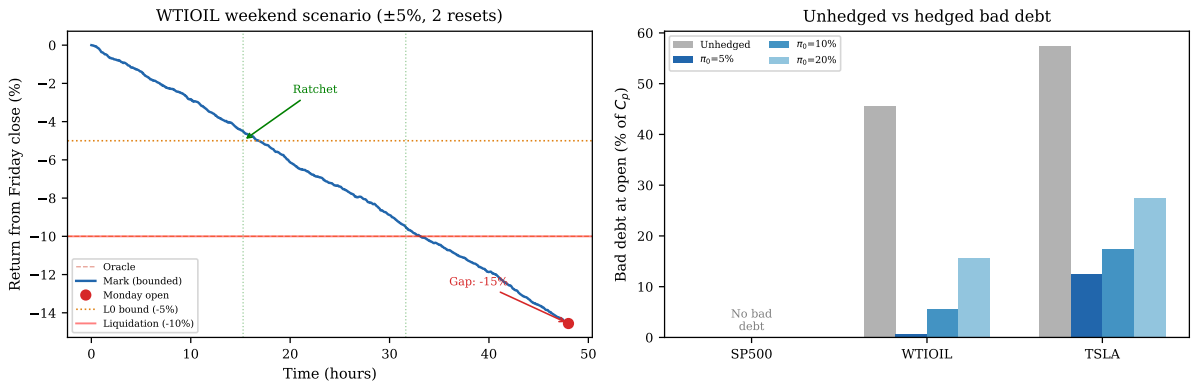

Figure 2: Gap hedging under Discovery Bounds v2 [DOC] + [SIM]. *Left:* WTIOIL scenario showing ratcheting and Monday-open snap beyond the liquidation threshold ($x_p = 10$). *Right:* Bad debt at open, unhedged vs. hedged at three buy prices with hedge notional $N_o = 0.5\,C_p$ (partial hedge). SP500: no bad debt (bound protective). WTIOIL and TSLA: hedges reduce bad debt by 52–99% depending on buy price and gap severity.

Two cases. First, at the Level 0 bound with full-notional hedging ($N_o = C_p$), the put triggers and the payout covers the perp loss, eliminating protocol-level bad debt. This is a mechanical consequence of the payoff structure [DOC]. Second, beyond Level 0, Figure 3 shows the more conservative scenario: a half-notional hedge ($N_o = 0.5\,C_p$) against gaps that exceed the bound through ratcheting or Monday-open snap. In this regime, hedges reduce but do not eliminate bad debt [SIM].

After re-anchoring (Section 2), the lower bound shifts below the original liquidation price, but a binary put struck at the initial liquidation threshold continues to protect because its strike is defined relative to entry price, not the active bound.

Evidence: Level 0 elimination is mechanical [DOC]; beyond-Level 0 reduction is simulated [SIM].

5.1 Break-Even Analysis

The binary put has positive expected value iff $P_{\text{gap}} \geq \pi_0/(1-\pi_0)$. Maximum profitable buy price: $\pi_0^* = P_{\text{gap}}/(1+P_{\text{gap}})$.

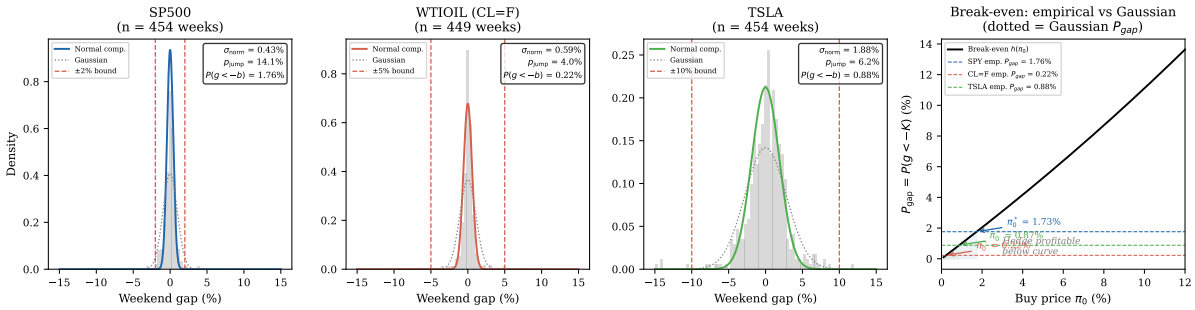

Figure 3: Break-even [EMP]. CL=F: $\pi_0^* \approx 0.22\%$; SPY: $\approx$1.73%; TSLA: $\approx$0.87%. Fat tails raise P_{gap} 2–5$\times$ above Gaussian, but absolute levels are low.

Binary puts at the liquidation threshold are instruments for elevated-risk scenarios, not routine weekend insurance [EMP].

5.2 Structured Payoff Profiles

Combining binary contracts with perpetuals within one HyperCore account produces four canonical strategies: protected long, capped long, zero-cost collar ($N_c = N_o^* \cdot \pi_{0,l}/\pi_{0,u}$), and range bet (Figure 4). All legs execute atomically within one block, eliminating the leg risk of multi-order construction on a CEX.

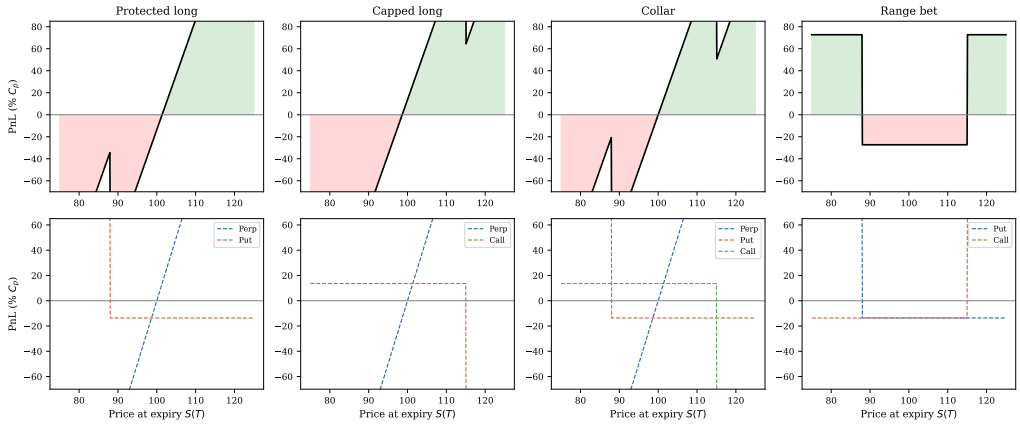

Figure 4: Structured payoffs [DOC]. Parameters: $K_l = 88\%$, $K_u = 115\%$, $\pi_0 = 12\%$.

6 Experiment 4: Weekend Funding Variance Hedging

A trader can suffer material funding costs over a weekend even if the Monday open gap is small, provided the crowd imbalance during the session is sustained. Conversely, a large gap can occur without generating extreme funding if the imbalance reverses quickly under the $\tau = 1$h oracle. This risk channel is distinct from gap risk (Experiment 3). A funding binary at a cumulative-funding threshold converts this open-ended exposure into a bounded one: in simulation, it reduces the standard deviation of weekend funding P&L by 10%.

The cost of that protection depends heavily on regime assumptions. Simulated $N = 5{,}000$ weekends with three regimes [SIM]:

Regime	Weight	Gap σ	Crowd bias	Noise
Normal	70%	0.5%	0.05%	0.04%
Event	20%	4%	0.8%	0.3%
Crisis	10%	10%	3.0%	0.8%

Binary contract: pays if cumulative funding $\mathcal{F} > 0.2\%$.

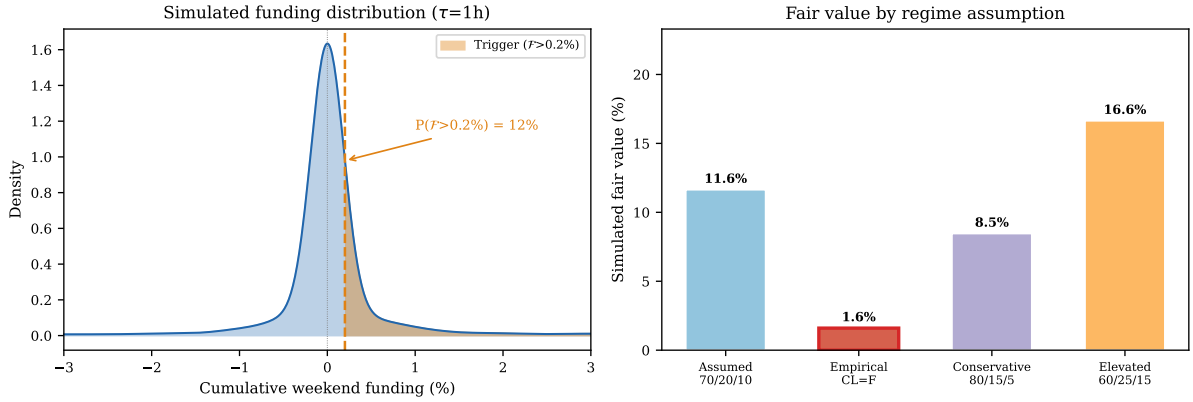

Figure 5: Funding binary [SIM]. *Left:* Simulated funding distribution with trigger region shaded. *Right:* Model fair value across four regime-weight scenarios. Empirical CL=F scenario highlighted.

Scenario	Weights	Fair value
Assumed (70/20/10)	70/20/10%	≈11.6%
Empirical CL=F	**96/3.1/0.9%**	**≈1.6%**
Conservative (80/15/5)	80/15/5%	≈8.5%
Elevated (60/25/15)	60/25/15%	≈16.6%

Fair values range from 1.6% to 16.6%, a 10× spread. This is the central result of the experiment: pricing a funding binary is equivalent to taking a view on crisis probability. The empirical CL=F scenario (96/3/1 weights, anchored to the observed bound-breach rate of 0.9%, i.e. 4 events in 449 weeks [EMP]) produces a fair value of 1.6%. The assumed scenario (70/20/10, a much heavier crisis tail) produces 11.6%. A trader or market maker who believes weekend tail risk is closer to the empirical distribution will price the instrument cheaply; one who believes the historical sample underestimates future crisis frequency will price it closer to 16.6%.

This means the eventual market price of funding binaries on HIP-4 will function as an implied crisis probability. A price of 5% implies the market assigns roughly 5% probability to the funding threshold being breached, which in turn implies a specific view on the mixture of normal, event, and crisis weekends. The within-regime parameters (gap σ, crowd bias, noise) are chosen assumptions, not fitted to on-chain data, so the absolute fair values carry model uncertainty.

Evidence: bound-breach frequency is empirical [EMP]; fair values are simulated [SIM].

6.1 Orthogonality of Price and Funding Binaries

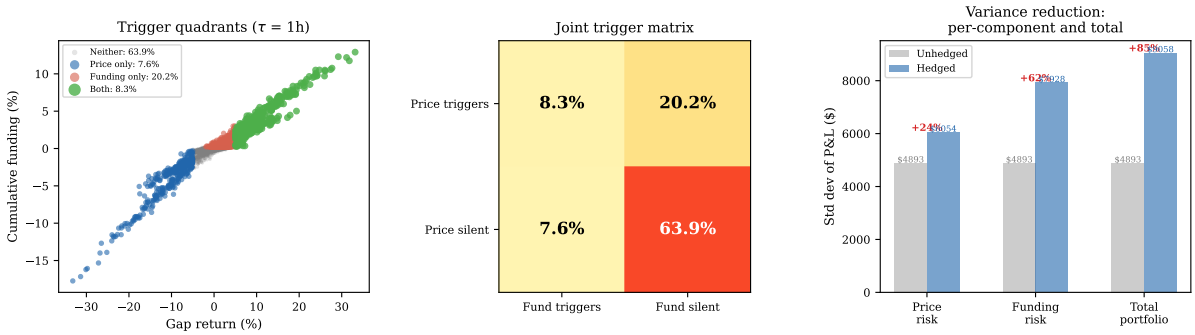

Figure 6: Risk-dimension analysis [SIM]. Co-trigger rate near zero. Per-component and total portfolio variance reduction.

The two instruments fire in largely non-overlapping regions. Per component, each reduces the relevant risk dimension's P&L standard deviation (price: −21%, funding: −10%). At the total-portfolio level, stacking both hedges *increases* standard deviation by ≈7%, because hedge premiums on non-trigger weekends are a net cost that adds dispersion. *Evidence:* simulated [SIM].

7 Counterparty Structure

The preceding experiments analyse binary hedges from the buyer's perspective. Whether the hedging thesis is viable depends on sellers willing to quote at prices near the break-even thresholds in Experiment 3.

7.1 Short Perp Holders

The risk channels in Experiments 3 and 4 are symmetric. A short perp holder profits when the gap put triggers, since the same downside move that pays the binary also benefits the short position. Selling puts is therefore a partial hedge against upside gap risk, funded by the premium. For funding binaries the logic is similar but operates through cash flows: shorts collect funding when the crowd is net-long, so a funding binary payout in that scenario partially offsets income the short was already receiving.

7.2 Market Makers and Vault Operators

HIP-3 vault operators carry weekend inventory risk regardless of whether they sell binaries. Quoting binary puts or funding binaries when they assess crisis probability as low lets them collect premium on tail exposure they already carry. Without binary markets, that exposure sits on their books uncompensated.

7.3 Protocol Insurance Fund

The HLP insurance fund absorbs bad debt from gap events without collecting any premium. Selling binary puts at actuarially fair prices would let the fund collect premium on quiet weekends and pay out on trigger weekends, growing the reserve over time instead of depleting it episodically. This requires governance specification of notional limits, pricing bands, and reserve requirements; the point here is narrower: the fund already bears the downside and currently receives nothing for it.

Evidence: structural [DOC]. Natural sellers exist, but their presence does not guarantee sufficient liquidity at break-even prices.

8 Limitations

The simulations assume no strategic behaviour, simplified orderbooks, and single-asset liquidation. These assumptions bias bad-debt estimates downward and hedge-effectiveness estimates upward.

Gap distributions are Yahoo Finance proxies for traditional-market weekend gaps. XYZ on-chain weekend price discovery operates under bounded mark prices with EWMA convergence, which may produce systematically different gap profiles. The proxy captures magnitude but not the path dynamics imposed by the relayer clamp and re-anchoring mechanism.

Funding dynamics use assumed crowd-bias and noise parameters (Section 6) that are not fitted to observed on-chain funding data. The within-regime parameters produce plausible outputs but are not calibrated; the regime *weights* for the empirical CL=F scenario are anchored to observed bound-breach frequencies, but the gap σ, crowd bias, and noise within each regime are chosen assumptions.

Binary instruments are assumed to be available at the stated prices with sufficient liquidity. The counterparty analysis (Section 7) establishes incentive alignment but not market depth; in practice, binary markets on HIP-4 may be thin, especially for tail-risk instruments with low break-even prices.

9 Governance Proposal: Portfolio Margin Extension

Problem. Most non-MAG7 XYZ markets remain `strictIsolated`, blocking the 7–14% capital efficiency gain for those assets.

Mechanism. When a HIP-4 put is registered as a hedge against a `strictIsolated` HIP-3 perp in the same account, the margin engine recognises the delta offset as if both were in cross-margin mode. Eligibility: same margin account, registered at entry with linked perp ID, outcome notional $\leq N_o^*$, same underlying. For MAG7 equities, which already support cross-margin, this extension is unnecessary.

Rationale. The isolated constraint prevents cross-position contagion. A binary put registered against a specific isolated perp adds no contagion risk: its payoff is bounded and its delta offset is confined to the isolated account.

10 Conclusion

Table 1: Robustness taxonomy of major findings.

Finding	Evidence	Main sensitivity	Status
1: Settlement window	[DOC]	None (mechanical)	Deterministic
2: Capital efficiency	[SIM]	Undocumented delta treatment	Conditional
3a: L0 bad-debt elim.	[DOC]	Payoff structure	Mechanical
3b: Beyond-L0 reduction	[SIM]	Hedge notional, gap magnitude	Illustrative
4: Funding binary fair values	[SIM]	Regime weights; crowd-bias params	Scenario-dependent

10.1 For deployers

The minimal product is a binary market paired with each weekend perp. The more interesting product is a bundled one: a "weekend-protected perp" where the frontend opens the leveraged

position and the corresponding funding binary (or gap put) atomically from the same margin account. The user sees one trade; the backend places two positions within HIP-4's block-finality window. This removes the friction of requiring traders to understand binary pricing and separately manage hedge positions.

Whether the market is viable depends on liquidity. Short holders and vault operators have structural reasons to sell (Section 7), and the spread between the modelled fair values (1.6–16.6%) and what buyers may be willing to pay for tail-risk protection leaves room for a market. But a binary market with three participants on each side is not a market. Deployers should expect to seed initial liquidity and accept that volumes will be thin until weekend perp open interest grows.

Two hard requirements: HIP-4 block-finality for settlement (the HIP-3 relayer clamp leaks up to 12.6% per event), and the portfolio margin extension to `strictIsolated` markets. Without the latter, capital efficiency gains are restricted to the seven MAG7 equities.

10.2 For traders

The funding binary is the more actionable of the two instruments. Its primary function is variance reduction on weekend funding costs: in simulation, it lowers the standard deviation of funding P&L by 10%. For a trader holding a leveraged position through 48 hours of internal pricing, that is the difference between a predictable cost and an open-ended one. The modelled fair values (1.6–16.6% depending on regime assumptions) provide a reference for evaluating quoted prices once liquid markets exist. If you are short, the same asymmetry works in reverse: you already collect funding when the crowd is long, so selling a funding binary is compensated by cash flows you would receive anyway.

Gap puts are a different story. Break-even prices sit below 2% for all three assets studied. At those levels, they only make sense ahead of weekends where you have a specific reason to expect an abnormal gap (earnings, macro events, geopolitical risk). As routine weekend insurance, the cost does not justify the protection for most position sizes.

One thing to avoid: buying both hedges together. Stacking a gap put and a funding binary increases portfolio variance by ≈7%, because the premiums on quiet weekends (the majority) accumulate as a net cost. The two instruments hedge different risks; pick whichever matches your exposure.